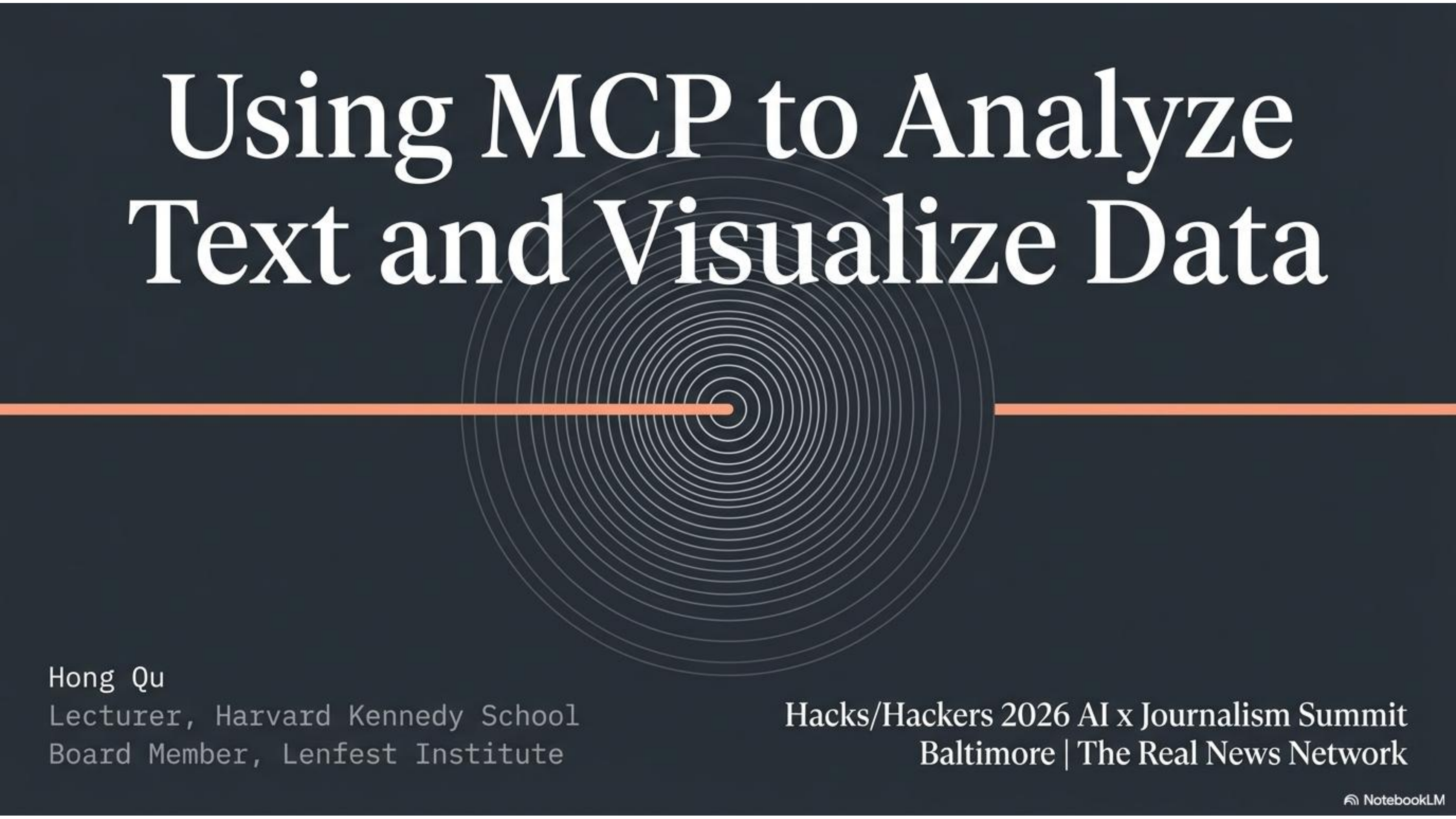


Using MCP to Analyze Text and Visualize Data

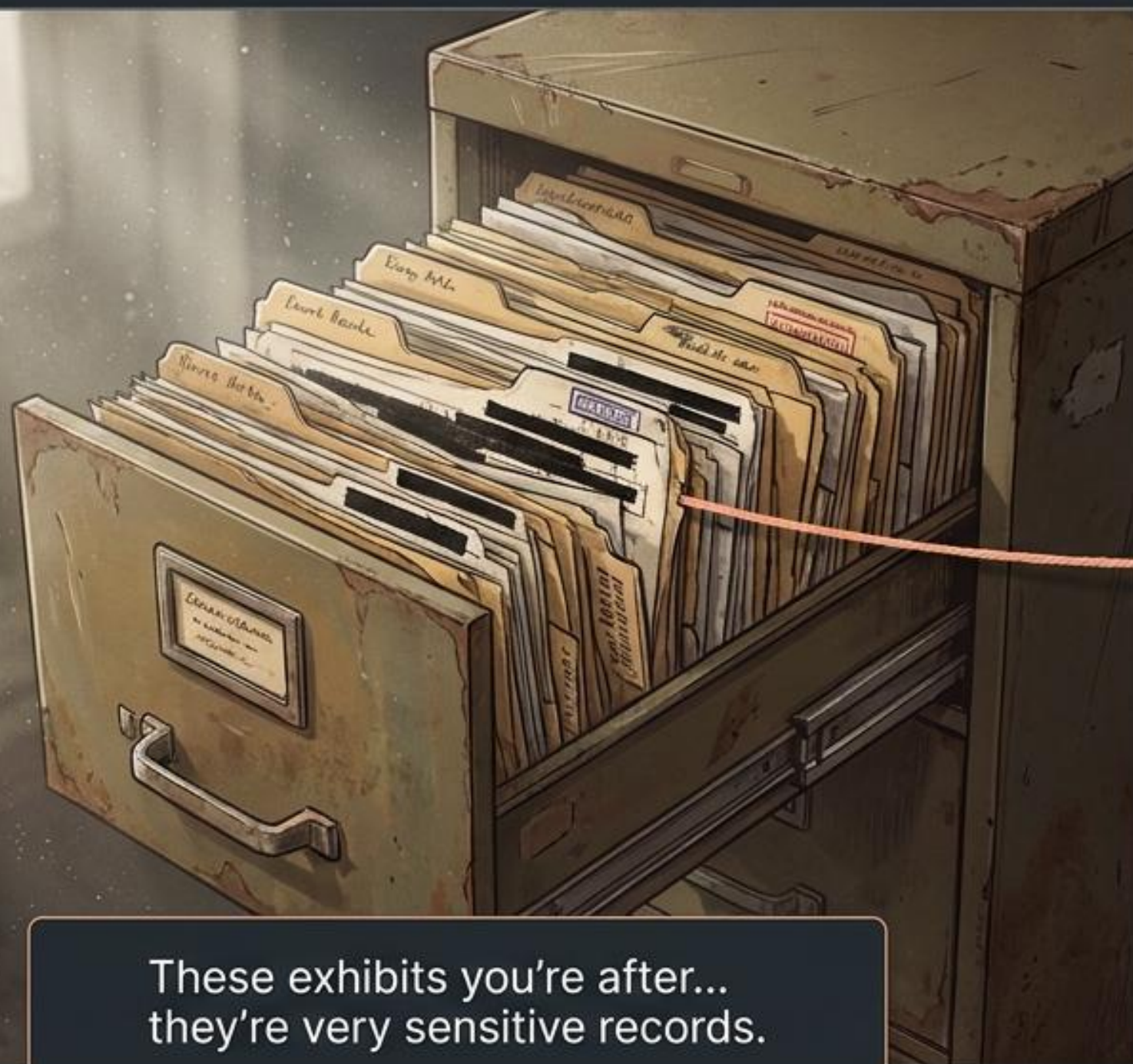
A decorative background graphic featuring a series of concentric circles centered in the middle of the slide. A horizontal orange line passes through the center of these circles, extending across the width of the slide.

Hong Qu

Lecturer, Harvard Kennedy School
Board Member, Lenfest Institute

Hacks/Hackers 2026 AI x Journalism Summit
Baltimore | The Real News Network

The investigative baseline requires building the database



These exhibits you're after...
they're very sensitive records.

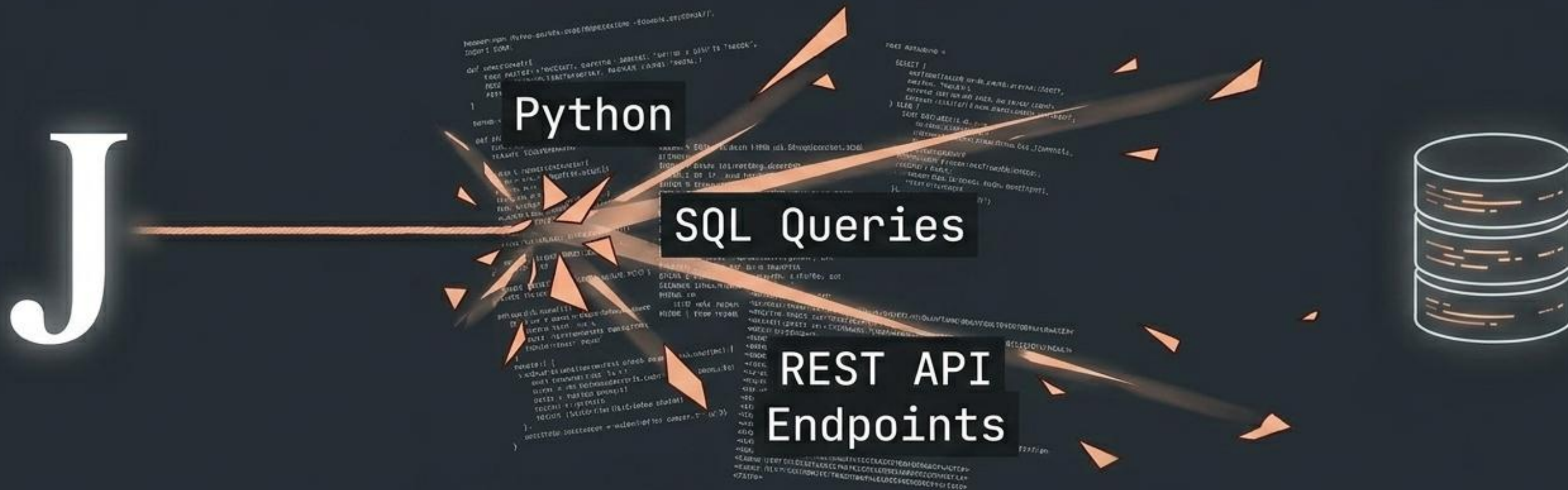
	A	B	C	D	E	F	G	H	I	J
	DATE	PRIEST_ID	LOCATION	ASSIGNMENT	STATUS	NOTES	TRANS_DATE	SOURCE	VERIFIED	LINK
001 002	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1970-08-01	Directory 78	YES	REF-SSA
003 003	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1970-08-01	Directory 78	YES	REF-SSA
004 005	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1978-06-01	Directory 78	YES	REF-SSA
025 036	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1978-05-21	Directory 78	YES	REF-SSA
037 038	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1979-05-12	Directory 78	YES	REF-SSA
029	1974-05-12	P-1824	Boston, MA	Active	Transferred - Pope John Inquiry (1.00120)	[REDACTED] Completed received.	1976-08-01	Directory 78	YES	REF-SSA
039 025	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1979-05-01	Directory 78	YES	REF-SSA
025 025	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1976-05-01	Directory 78	YES	REF-SSA
036 036	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1978-03-05	Directory 78	YES	REF-SSA
036 031	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1978-05-22	Directory 78	YES	REF-SSA
028 028	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1976-03-01	Directory 78	YES	REF-SSA
036 035	1974-05-12	P-1824	Boston, MA	St. Mary's Parish Completed received.	Active		1976-03-01	Directory 78	YES	REF-SSA
056 065	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1978-03-12	Directory 78	YES	REF-SSA
086 087	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1978-03-12	Directory 78	YES	REF-SSA
066 068	1974-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1978-03-12	Directory 78	YES	REF-SSA
000 100	1976-05-12	P-1824	Boston, MA	St. Mary's Parish	Active		1978-03-12	Directory 78	YES	REF-SSA

The Spotlight team painstakingly extracted transferred priest records from directories to build a manual database.

The lesson: Journalists had to learn to use a spreadsheet.

The modern data access chasm

Today's investigations require querying millions of records, but the data is locked behind APIs requiring strict syntax and programming expertise.



The journalist's dilemma: You know the story, but you don't speak the machine's language.

Bridging the gap with Model Context Protocol

A universal, open standard for connecting AI systems with data sources, replacing fragmented integrations with a single protocol. — **Anthropic**



**No Code. No SQL.
No Python.**

Two-way, secure
connections between
external tools
and AI agents.

Bypasses information
silos and legacy systems
instantly.

The shift from programmatic access to agentic access

	Traditional API	MCP Server
Interaction Paradigm	Speak Machine: Rigid parameters, strict endpoints.	Speak Librarian: Conversational, natural language queries.
User Requirement	Developer-focused: Requires custom connector code.	Journalist-focused: Zero code, universal standard.
Output Handling	Returns raw JSON; user must manually parse and chart.	Agent executes next steps: downloads dockets, builds visualizations, maps networks.

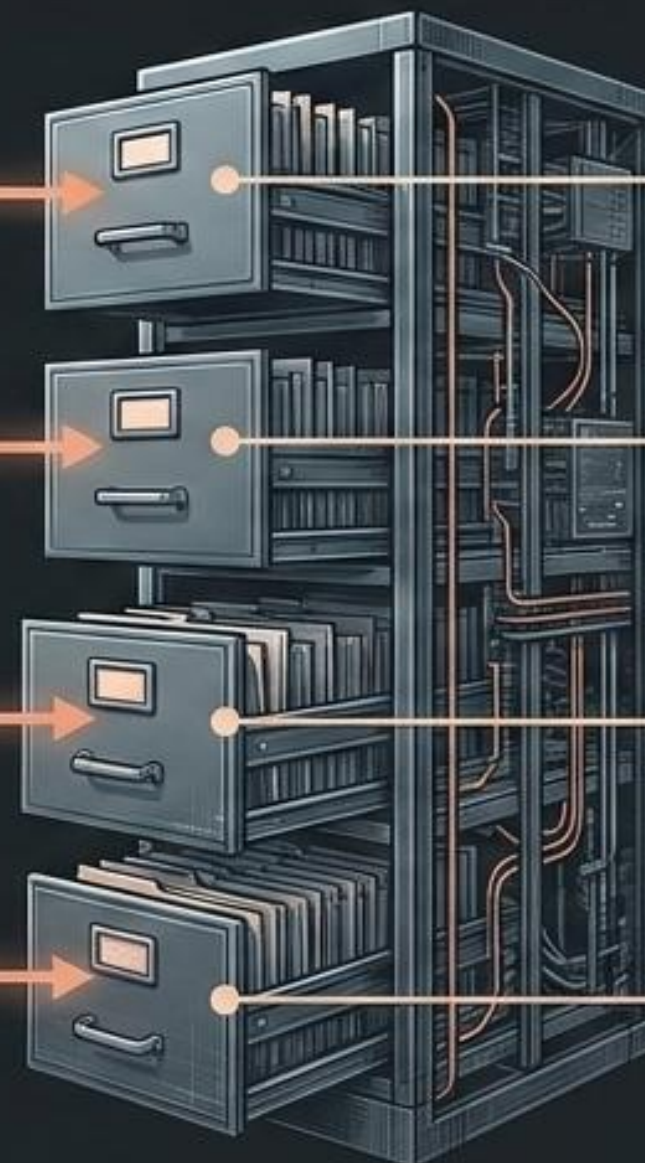
Agentic access to public records via CourtListener

Free Law Project Archives



User Prompt

Claude



- Millions of federal and state case laws and opinions.
- PACER data: cases, parties, attorneys, dockets.
- Judicial financial disclosures: assets and debts triggering conflicts.
- Searchable audio and transcripts of oral arguments.

Talk to the legal archives



I want to get all the latest documents for the OpenAI case involving Sam Altman and Elon Musk.

Connecting to mcp.courtlistener.com...
Accessing Docket 69013420: Musk v. Altman...
> █

The AI assistant bypasses standard web search, queries the live PACER database, retrieves the verified docket, and presents the latest filings grounded in source documents.



Generate a network visualization for this case's citations. AI maps what cases cite it, and what it cites.



Verify every citation in this brief. The agent runs grounded citation checks against the live archive to eliminate hallucinations.



Monitor real-time filings, citations, and queries, setting up alerts for specific dockets.

The business case for hosting your own MCP

You don't have to release your raw data.
You rent AI access to it.



Protect the raw asset

Like an API, the underlying dataset remains secure and proprietary.



Control functionality

Limit access to specific tools, endpoints, or data slices based on user permissions.



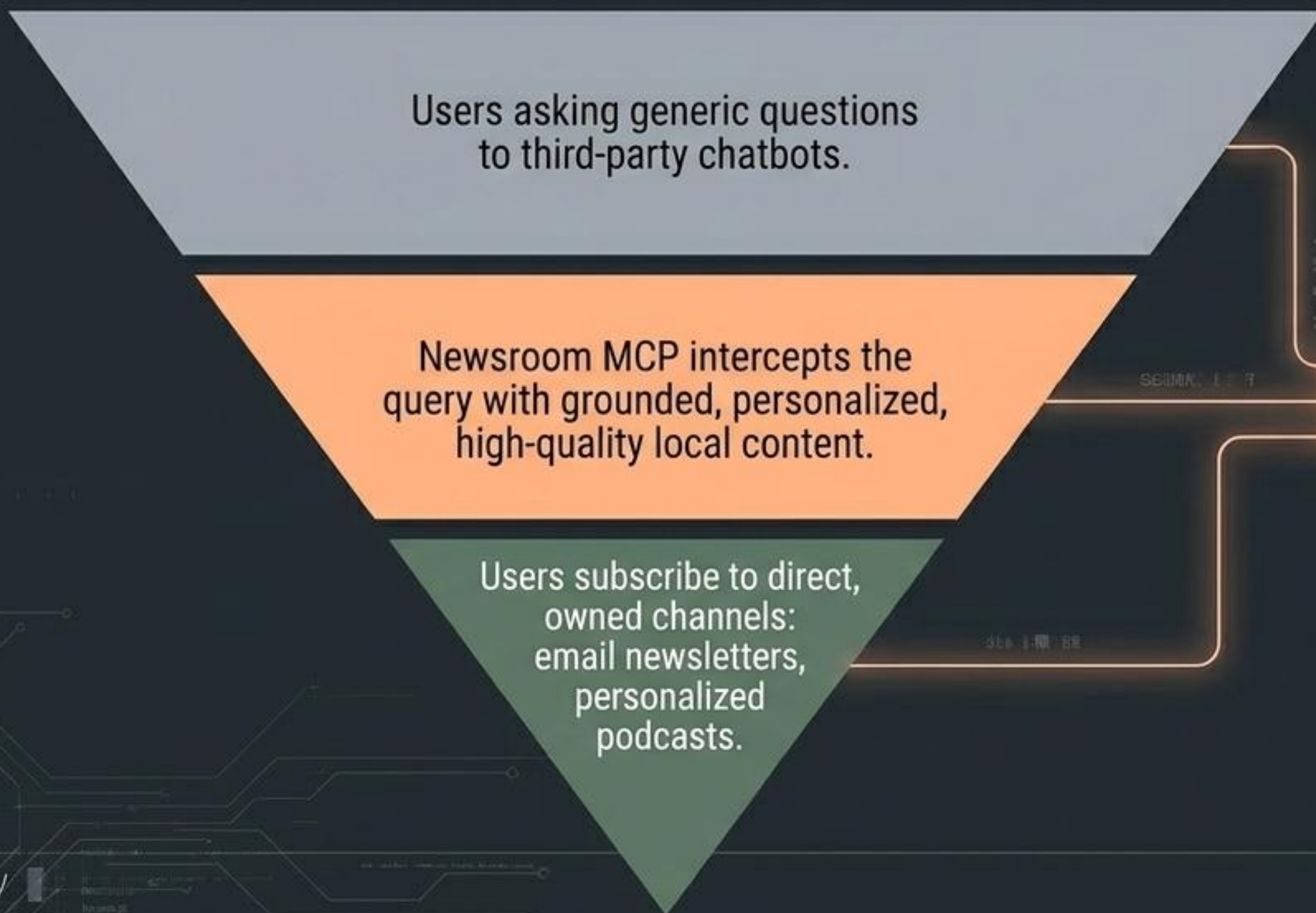
Empower the audience

Allow customers to query your journalism-grounded data on demand using the chatbots they already know.

Fulfilling local needs with Concierge Journalism



Owning the audience relationship



The 100x Multiplier

Intercept users before they turn to hallucination-prone chatbots. Using an MCP to push personalized content directly to your audience builds an owned relationship that operates at a 100x multiplier over passive reading.

The quantitative challenge

How do we make sense of this without writing code?

```
raw.githubusercontent.com/.../myDatasauRus.csv
142 rows of raw X/Y coordinates
```


From raw text to instant insight



Upload the raw data file.
Prompt 1: I want to make a scatterplot using dino.

Visualizes the distinct dataset shape.

Prompt 2: Make a scatter plot for all the datasets and put them into a grid of scatterplots.

Returns comparative summary statistics for away, bullseye, dino, and star datasets.

Building an API using natural language

1. Do you see the csv file?

2. I want to create an API server and host it on GitHub Pages.

3. The API has 1 function: users can request to get all the x and y values for a specific dataset by sending the dataset name.

4. Push this code to my github repository named MCP_demo_datasaurus.

Result: The AI writes the server code and structures the static JSON endpoints for GitHub Pages without requiring human engineering.

```
api_server.py

import pandas as pd
import json
from http.server import BaseHTTPRequestHandler, HTTPServer
from urllib.parse import urlparse, parse_qs

class RequestHandler(BaseHTTPRequestHandler):
    def do_GET(self):
        parsed_path = urlparse(self.path)
        query = parse_qs(parsed_path.query)
        dataset_name = query.get('dataset', [None])[0]

        if dataset_name:
            try:
                df = pd.read_csv('nyDatasauRus.csv')
                filtered_df = df[df['dataset'] == dataset_name]
                data = filtered_df[['x', 'y']].to_dict(orient='records')
                response = json.dumps(data)

                self.send_response(200)
                self.send_header('Content-type', 'application/json')
                self.end_headers()
                self.wfile.write(response.encode())
            except FileNotFoundError:
                self.send_response(500)
                self.end_headers()
                self.wfile.write(b'Dataset file not found.')
            except Exception as e:
                self.send_response(500)
                self.end_headers()
                self.wfile.write(str(e).encode())
        else:
            self.send_response(400)
            self.end_headers()
            self.wfile.write(b'Please provide a dataset name.')

    def run(server_class=HTTPServer, handler_class=RequestHandler, port=8000):
        server_address = ('', port)
        httpd = server_class(server_address, handler_class)
        print(f'Starting httpd on port {port}...')
        httpd.serve_forever()

if __name__ == '__main__':
    run()
```


Converting the API into an MCP server

```
GET /data
HTTP/1.1
Host: api.example.com

GET /data
...
```

In addition to the API, I want to make an MCP server to be hosted on fastmcp.cloud. Can you generate the code for me to upload there?

```
import FastMCP

@fastmcp.tool
import FastMCP

@fastmcp.tool
def list_datasets():
    res = list_datasets()
    return tool

def get_dataset_points(name: str):
    if get_dataset_points(name: str):
        return name()
    return dataset_points(name, name: str))

...
```

Available MCP tools generated:

- `list_datasets`: List dataset names and point counts.
- `get_dataset_points`: Get all x and y points.
- `get_dataset_summary`: Get count and x/y bounds.

comm > █

DATA PAGE 3004. 11.465A KLCR-29G:

> comm >

comm > █

github.com/hqu/MCP_demo_datasaurus

Deploying on FastMCP and connecting to Claude

1

Host the Server



Upload the generated code to fastmcp.cloud (Horizon Prefect).

2

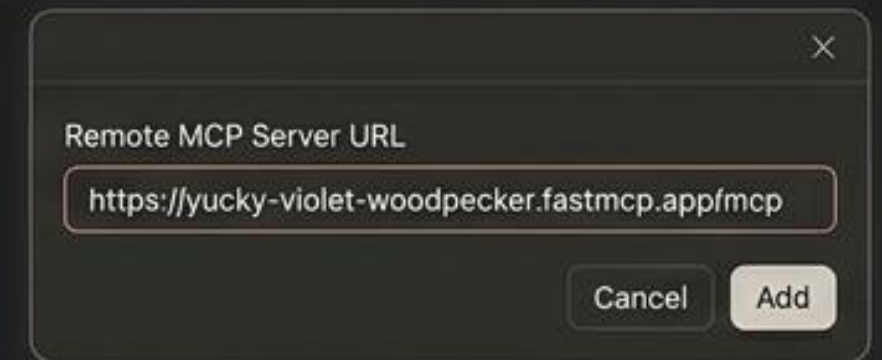
Add Custom Connector



Open Claude's connector settings (Settings > Customize). Click to add a new custom connector.

3

Paste the URL



Name the connector and paste the Remote MCP server URL. Click 'Add' to save.

Essential Resources & Repositories

Text Analysis (CourtListener)

CourtListener MCP Server
for Agentic Access



[wiki.free.law/c/courtlistener/...](https://wiki.free.law/c/courtlistener/)

Data Visualization (DatasauRus)

Hong Qu's GitHub Demo
Repository



github.com/hqu/MCP_demo_datasaurus

Open Standard (Anthropic)

Introducing the Model
Context Protocol



mcp.directory